

REINVENT-Based Generative Design of Factor Xa Inhibitors

Using Scaffold-Aware ML Scoring

Formal Project Write-Up Summary - Updated with Rate-Normalized Results

Abstract

This project developed an end-to-end generative molecular design workflow for Factor Xa inhibitor discovery using REINVENT4 transfer learning and a scaffold-aware RandomForest activity model from Project 1. A short 5-epoch transfer-learning run was performed using potent FXa seed molecules and evaluated against the original REINVENT prior using the same scoring model. Because the prior and transfer-learned model produced different numbers of valid molecules (982 vs 911), enrichment was assessed using both raw counts and rates. Transfer learning increased the fraction of generated molecules with predicted $pK_i \geq 7$ from 10.4% to 18.8% and generated 4 molecules with predicted $pK_i \geq 8$, compared with none from the prior. After novelty and drug-likeness filtering, 600 valid, novel, druglike TL candidates remained, including 92 candidates with predicted $pK_i \geq 7$ and a top-30 docking queue for structure-based follow-up.

1. Executive Summary

The main result is a clear, rate-normalized enrichment of high-scoring predicted FXa molecules after REINVENT transfer learning. The number of predicted $pK_i \geq 7$ molecules increased from 102 to 171, and the rate increased from 10.4% of valid prior-generated molecules to 18.8% of valid TL-generated molecules. The TL model also produced 4 predicted $pK_i \geq 8$ molecules, whereas the prior produced none.

The main trade-off was drug-likeness: the basic druglike rate decreased from 80.5% in the prior sample to 65.9% in the TL sample. This motivated the final candidate-prioritization stage, which focused on novel, druglike, high-scoring molecules rather than raw ML score alone.

2. Project Objective

The objective was to determine whether REINVENT transfer learning could shift a general molecular prior toward FXa inhibitor-like chemical space, and whether a scaffold-aware FXa ML model could quantify that shift. The project was designed as a practical computational drug discovery workflow connecting ligand-based ML scoring, generative design, and docking-ready candidate triage.

3. Methods

3.1 FXa Activity Model

Generated molecules were scored using the Project 1 scaffold-aware FXa RandomForest model saved as `models/fxa_05b_best_scaffold_feature_model.joblib`. The model pipeline consisted of SimpleImputer, VarianceThreshold, and RandomForestRegressor. Molecules were represented using binary Morgan fingerprints with radius 2 and 2048 bits.

3.2 Seed Preparation and Prior Compatibility Filtering

Potent FXa molecules with $pK_i \geq 8.0$ were selected as transfer-learning seeds. After flattening stereochemistry and deduplicating canonical SMILES, 1211 unique potent FXa SMILES remained. An RDKit element filter removed 12 molecules containing unsupported elements for the REINVENT prior, leaving 1199 compatible SMILES.

Filtering step	Count
Initial potent seed molecules	1249
Unique flat potent FXa SMILES	1211
REINVENT-compatible SMILES retained	1199
Removed due to unsupported elements	12
Removed elements	B: 9; I: 2; P: 1

3.3 Transfer Learning and Sampling

REINVENT4 was run in transfer-learning mode for a short CPU sanity run. The filtered all-flat FXa seed set was split into approximately 959 training SMILES and 240 validation SMILES. Transfer learning was run for 5 epochs using the official REINVENT prior. The best validation loss occurred at epoch 5, but the validation curve did not reach a clear minimum, so this run is treated as a proof-of-concept rather than a fully optimized production model.

Parameter	Value
run_type	transfer_learning
device	cpu
num_epochs	5
batch_size	50
sample_batch_size	1000
standardize_smiles	false
randomize_smiles	true
randomize_all_smiles	false

3.4 Scoring and Novelty Analysis

Both the original REINVENT prior and the FXa transfer-learned model were sampled for 1000 molecules using identical sampling settings. Generated molecules were parsed with RDKit, converted to flat canonical SMILES for novelty analysis, fingerprinted using the RDKit MorganGenerator, and scored with the Project 1 FXa model. RF uncertainty was estimated as the standard deviation across RandomForest tree predictions after applying the fitted preprocessing steps.

4. Results

4.1 Prior vs Transfer-Learned Model Comparison

Because the prior and TL model produced unequal valid molecule counts, raw counts alone are not sufficient. The table below reports both raw counts and rates normalized by the number of valid generated molecules.

Metric	Prior 1000	TL 1000	Interpretation
Valid generated molecules	982	911	TL produced fewer valid unique molecules after sampling.
Mean predicted pKi	6.531	6.674	+0.143 pKi shift.
Median predicted pKi	6.509	6.647	+0.138 pKi shift.
Max predicted pKi	7.922	8.942	+1.020 pKi increase.
Top-10 mean predicted pKi	7.573	8.021	+0.448 pKi enrichment among the best molecules.
Predicted pKi >= 7	102 / 982 (10.4%)	171 / 911 (18.8%)	+69 molecules; +8.4 percentage points.
Predicted pKi >= 8	0 / 982 (0.0%)	4 / 911 (0.4%)	TL produced high-scoring molecules absent from the prior sample.
Novel vs FXA reference	982 / 982 (100.0%)	911 / 911 (100.0%)	Both sets were fully novel by flat canonical SMILES comparison.
Basic druglike molecules	791 / 982 (80.5%)	600 / 911 (65.9%)	-191 raw count; -14.7 percentage-point decrease.

The central result is the rate-normalized increase in predicted high-scoring molecules. The fraction of valid molecules with predicted pKi >= 7 increased from 10.4% to 18.8%, a +8.4 percentage-point improvement. The TL model also produced 4 molecules with predicted pKi >= 8, whereas the prior produced none.

4.2 Candidate Prioritization

After novelty and drug-likeness filtering, 600 valid, novel, druglike TL-generated molecules remained. Among these 600 candidates, 92 had predicted pKi >= 7 and were assigned to Tier 1, Tier 2, or Tier 3 candidate groups.

Tier	Criteria	Count
Tier 1	Novel + druglike + predicted pKi >= 8.0	2
Tier 2	Novel + druglike + predicted pKi 7.5-8.0	13
Tier 3	Novel + druglike + predicted pKi 7.0-7.5	77
Total tiered candidates	Novel + druglike + predicted pKi >= 7.0	92
Docking queue	Top prioritized candidates for structure-based follow-up	30

4.3 Headline Candidate

The top prioritized candidate in the docking queue was selected after applying novelty, drug-likeness, predicted pKi, QED, and RF uncertainty considerations.

Property	Value
SMILES	<chem>O=c1cccn1-c1ccc(NS(=O)(=O)c2ccc(Cl)s2)cc1</chem>
Predicted FXA pKi	8.116
RF tree uncertainty	1.290
MW	366.85
cLogP	3.35
QED	0.769
Novel vs FXA reference	Yes
Basic druglike flag	Yes

5. Discussion

The transfer-learning result is scientifically meaningful because the enrichment remains clear after correcting for unequal valid molecule counts. The TL model did not merely produce more high-scoring molecules as a function of sample size; it produced a higher fraction of high-scoring molecules. The increase in top-10 mean predicted pKi and the appearance of pKi ≥ 8 candidates suggest that the model learned features associated with the potent FXa seed distribution.

The result should still be interpreted carefully. The TL model decreased the fraction of basic druglike molecules, and the generated molecules are outside the exact training reference set. Therefore, the model is useful for prioritization, but not sufficient for claiming activity. The correct next step is structure-based validation using the top-30 docking queue.

6. Limitations

- Predicted pKi values are model-based prioritization scores, not experimental affinities.
- The transfer-learning run was intentionally short and CPU-based, so it is a proof-of-concept sanity run.
- The RandomForest model has uncertainty, especially for novel generated molecules.
- The druglike fraction decreased after transfer learning, requiring downstream filtering.
- Docking, binding-mode analysis, ADMET filtering, and synthetic feasibility checks remain necessary.
- No experimental validation was performed.

7. Deliverables

Deliverable	Path
Final key metrics	results/tables/project2_final_key_metrics.csv
Summary JSON	results/metrics/project2_final_summary.json
Summary text	results/metrics/project2_final_summary.txt
Prior vs TL key metric figure	results/figures/prior_vs_tl_key_metrics.png
Predicted pKi distribution figure	results/figures/prior_vs_tl_predicted_pki_distribution.png
Candidate tier figure	results/figures/tl_candidate_tier_counts.png
Tier 1 candidate list	results/tables/tl_fxa_tier1_pki8_druglike_candidates.csv
Top-100 prioritized list	results/tables/tl_fxa_top100_ranked_candidates.csv
Docking queue	results/tables/tl_fxa_docking_queue_top30.csv

8. Conclusion

This project shows that REINVENT transfer learning can shift a general molecular prior toward FXa inhibitor-like chemical space when guided by potent FXa seeds. Compared with the original prior, the TL model increased the rate of predicted pKi ≥ 7 molecules from 10.4% to 18.8% and generated 4 predicted pKi ≥ 8 molecules. After novelty and drug-likeness filtering, the project produced 600 viable TL candidates and a top-30 docking queue. The project provides a professional bridge from scaffold-aware ligand-based ML to generative molecular design and structure-based validation.